

10 OVERCLOCKING TIPS

Now that your system is all set firmly, you may feel the urge to push its performance that extra mile. Here are a few things you need to keep in mind before embarking on any overclocking adventure

1 With a little help from your motherboard

Firstly, consult the manual to make sure that your motherboard supports overclocking through jumper settings or BIOS tweaks. Once you are sure that you can change the CPU's core voltage, clock multiplier or bus speed, only then should you go about pushing your system harder.

2 Tighter control of voltage/frequencies

Opt for a motherboard that offers fine control of parameters such as the core voltage, clock multiplier and bus frequency. If these parameters can be incremented by small units, your overclocking experiments can be that much more controlled and safer.

3 Processor permission

Most processors allow overclocking only to a certain extent. Others, by virtue of their design, allow overclocking to greater degrees. A processor with a lower amount of L2 cache, a smaller fabrication process and a lower bus speed support can be overclocked to far greater extents as it has more room for expansion.

Most processors today have their clock multipliers locked and this prevents

them from being tweaked through that component. In such cases, the bus speed needs to be used, but take care so that the rest of the system is not adversely affected.

4 You are on your own!

Keep in mind that your warranties will not cover any damage cause by tweaking; so be very sure of the extent to which you can push your system safely. This is especially applicable if you want to modify any part of your hardware to accommodate experiments.

5 Ramp up your cooling

The first by-product of a tweaking experiment will be the generation of heat in your critical components. Therefore, make sure that you

have adequate cooling for components such as the processor, RAM and motherboard chipset. If necessary, purchase a better heatsink or cabinet cooling system.

6 Multiplier or bus speed?

You can increase the effective speed of your system by either increasing the clock multiplier or the bus frequency. However, since most processors are clock-locked, you will have to take the bus-frequency route. However, in this case, installing a system monitoring software would be a good idea to keep tabs on your temperatures, voltages and frequencies. See that limits are not exceeded—temperatures should not exceed 60° C and voltages should not swing more than 5 per cent away from their rated values. Throttle back as soon as you feel that your system is reaching its limits and if it shows any signs of instability.

7 Overclock in steps

Start out your overclocking at the rated settings and work your way up in small increments—do not be overzealous. Working in small steps allows you to closely monitor your system for signs of instability, before pushing it to the next level.

8 Tabulate your findings and find out the most stable setting for your system

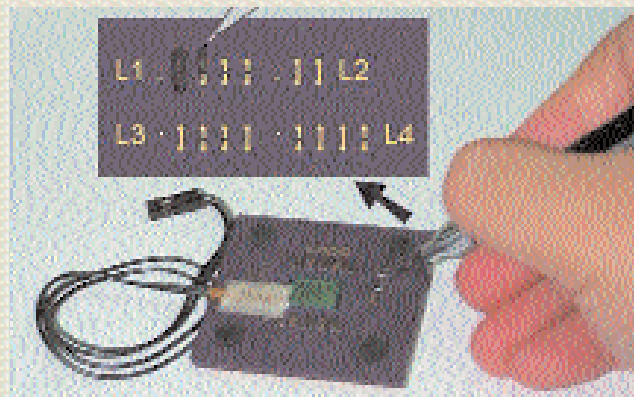
At each step of your overclocking adventure, tabulate the present frequency, voltage and thermal levels of your processor and system. At each stage, run a strenuous application such as a looping game benchmark and look for any signs of instability. Your tabulated results will pin down the point at which your system runs most stable.

9 Monitor the status of your system

Throughout your overclocking efforts, monitor your system temperatures and voltages very closely. It is critical that they remain within limits for your system to run reliably. Failing to do so will result in shortening the life of your processor or even failure.

10 Run an acid test

Once you've overclocked to satisfaction, you need to run a burn-in application for at least 4 hours. This will ensure that your overclocked computer will run all your applications reliably, without lag or failing.



Some AMD processors require their L1 bridges to be shorted with a graphite pencil before overclocking them